Model	Correct	Not attempted	Incorrect	Correct given attempted	F-score
Claude-3-haiku (2024-03-07)	5.1	75.3	19.6	20.6	8.2
Claude-3-sonnet (2024-02-29)	5.7	75.0	19.3	22.9	9.2
Claude-3-opus (2024-02-29)	23.5	39.6	36.9	38.8	29.3
Claude-3.5-sonnet (2024-06-20)	28.9	35.0	36.1	44.5	35.0
GPT-4o-mini	8.6	0.9	90.5	8.7	8.6
GPT-4o	38.2	1.0	60.8	38.0	38.4
OpenAI o1-mini	8.1	28.5	63.4	11.3	9.4
OpenAI o1-preview	42.7	9.2	48.1	47.0	44.8

Table 3: Performance of various models on SimpleQA. F-score is the harmonic mean between correct and correct given attempted; see Appendix B for discussion.

4 Measuring calibration

A factuality benchmark like SimpleQA allows us to measure the scientific phenomenon known as calibration, or whether language models "know what they know." One way to measure calibration is to directly ask the language model to state its confidence in its answer using a prompt like: "Please give your best guess, along with your confidence as a percentage that that is the correct answer" (for exact prompt, see Appendix C). Then we can plot the correlation between the stated confidence of the model, and how accurate the model actually was. A perfectly calibrated model would have the same actual accuracy as stated confidence. For instance, on all prompts where the model stated a confidence of 75%, the accuracy would be 75% for a perfectly calibrated model.

This result is shown in Figure 2 (left). The positive correlation between stated confidence and accuracy is a reassuring sign that models have some notion of confidence. We see that o1-preview is more calibrated than o1-mini, and gpt4o is more calibrated than gpt4o-mini, which is consistent with prior work showing that larger models are more calibrated. However, the fact that performance is well below the line $y = x$ means that models consistently overstate their confidence. Hence, there is a lot of room to improve the calibration of large language models in terms of stated confidence.

Another way to measure calibration is to ask the language model the same question 100 times (here, we use temperature 1). Since language models may produce different answers upon repeated attempts, we can assess whether frequency of an answer corresponds to its correctness. Higher frequency typically indicates that the model is more confident in its answers, as the model is giving the same answer repeatedly. A calibrated model would have the same accuracy as answer frequency.

In the Figure 2 (right), we show the calibration of language models as measured by the frequency of their responses. Here we use string match to group together different answers from the language model using the same prompt as the stated confidence figure. For each question, we only consider the most-frequent answer. We see across all models that accuracy increases with frequency, and that o1-preview has the highest level of calibration, where the frequency of the response is roughly equivalent to the accuracy of the response. Similar to calibration via stated confidence plot above, we again see o1-preview is more calibrated than o1-mini, and gpt4o is more calibrated than gpt4o-mini.

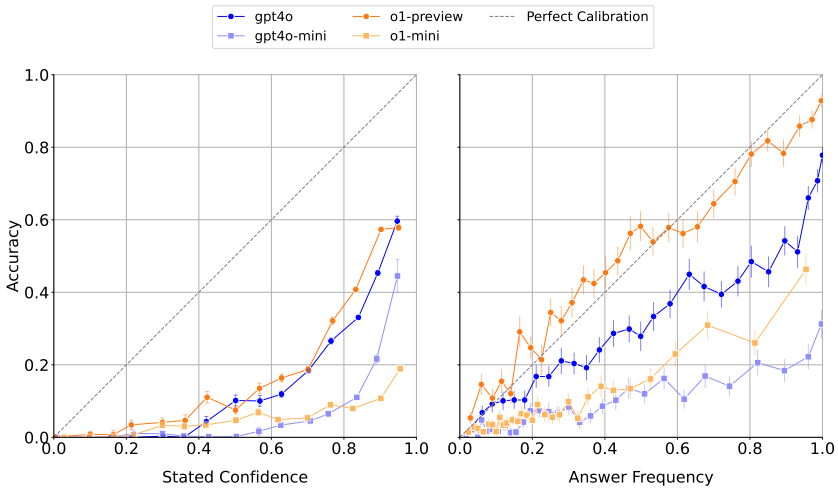


Figure 2: Left: Calibration of language models based on their stated confidence, uniformly binned into 15 intervals. Right: Calibration assessed by asking models the same question 100 times, quantile-binned into 30 intervals.

5 Related work and discussion

In this paper we have proposed a very simple benchmark for measuring the factuality of language models. SimpleQA follows several prior benchmarks that aim to measure the ability of language models to provide knowledge about the world. Perhaps the two most similar benchmarks are TriviaQA (Joshi et al., 2017) and Natural Questions (Kwiatkowski et al., 2019), which were good datasets at the time but are too easy for today's language models. Other recent and related benchmarks include LongFact (Wei et al., 2024), a benchmark of open-ended prompts, and FreshQA (Vu et al., 2023), a QA benchmark to evaluate performance on fast-changing knowledge, among other work (Lin et al., 2022a; Li et al., 2023; Cheng et al., 2023; Min et al., 2023; Zhao et al., 2024; Krishna et al., 2024).

In our brief experiments we also measured calibration in language models, building on a long history of prior work studying whether neural nets are calibrated. Notably, Kadavath et al. (2022) studied calibration in language models, finding that they were increasingly calibrated as a function of training compute, where calibration is measured by the having the language model give the probability of a true statement. Our finding that frequency of answer correlates with accuracy is also consistent with Wang et al. (2023), which found a similar result for a different model on a math word problem benchmark. Other work has attempted to improve the calibration of language models (Lin et al., 2022b; Agrawal et al., 2024), which could lead to increased adoption of language models in real-world settings.

A main limitation with SimpleQA is that while it is accurate, it only measures factuality under the constrained setting of short, fact-seeking queries with a single, verifiable answer. Whether the ability to provide factual short answers correlates with the ability to write lengthy responses filled with numerous facts remains an open research question. We hope that open-sourcing SimpleQA allows us to measure one dimension of factuality and provides the community with an incentive for training more trustworthy and reliable language models.

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